Antibiotics and Antimicrobial Resistance

Knowledge document 050 of 100

Introduction to Antibiotics

Antibiotics are medicines that kill bacteria or stop them from multiplying, and they are used to treat bacterial infections such as pneumonia, typhoid, tuberculosis, urinary tract infections and strep throat. Before antibiotics, even small wounds or common infections could be fatal; in the early twentieth century, bacterial diseases were a leading cause of death. The discovery of antibiotics transformed medicine and made surgery, cancer chemotherapy, organ transplants and childbirth much safer. However, the misuse of these precious drugs has produced a global threat called antimicrobial resistance (AMR), which could make many infections untreatable again. The World Health Organization considers AMR one of the top global public health threats.

Who Discovered Penicillin?

Penicillin was discovered by the Scottish scientist Alexander Fleming in 1928 at St. Mary's Hospital in London. After returning from a holiday, he noticed that a culture dish of Staphylococcus bacteria had been contaminated by a mould, Penicillium notatum, and that bacteria around the mould had been killed. He named the active substance penicillin. The substance was difficult to produce, and it was Howard Florey and Ernst Chain, with Norman Heatley, at Oxford University who developed it into a usable drug in the early 1940s. Mass production in the United States during World War II saved countless soldiers' lives. Fleming, Florey and Chain shared the 1945 Nobel Prize in Physiology or Medicine. Earlier, Paul Ehrlich had developed Salvarsan in 1909 to treat syphilis, and later antibiotics include streptomycin, discovered in 1943 by Selman Waksman and used against tuberculosis, tetracycline, erythromycin and many others. In his Nobel lecture, Fleming warned that careless use could lead to resistant bacteria.

How Antibiotics Work

Antibiotics target features unique to bacteria. Some, like penicillin and cephalosporins, block the building of the bacterial cell wall, causing bacteria to burst. Others, such as tetracyclines and macrolides, block the bacterial ribosomes and stop protein making. Some, such as fluoroquinolones, stop bacterial DNA copying. Antibiotics are bactericidal (kill bacteria) or bacteriostatic (stop growth). They are classified into groups, such as penicillins, cephalosporins, macrolides, tetracyclines, aminoglycosides and quinolones.

Why Don't Antibiotics Work on Viruses?

Antibiotics do not work on viruses because viruses are completely different from bacteria. Bacteria are living single-celled organisms with their own cell wall and machinery for making proteins and copying DNA, and antibiotics attack those structures. Viruses are much smaller, are not cells, have no cell wall or ribosomes, and can only multiply by entering the cells of a host and using the host's machinery. There is therefore nothing in a virus for an antibiotic to attack. Illnesses caused by viruses, such as the common cold, influenza, COVID-19, dengue, most sore throats and most coughs, do not respond to antibiotics. Taking them for such illnesses does not help recovery, can cause side effects such as diarrhoea and allergic reactions, and contributes to resistance. Viral infections are prevented by vaccines and sometimes treated with specific antiviral drugs, such as oseltamivir for influenza or acyclovir for herpes, while many resolve with rest and fluids.

What Is Antimicrobial Resistance?

Antimicrobial resistance occurs when bacteria, viruses, fungi or parasites change over time and no longer respond to the medicines designed to kill them. When this concerns bacteria and antibiotics it is called antibiotic resistance. The bacteria are not the person's body becoming resistant; it is the germs that become resistant. Resistant bacteria are often called superbugs. Examples include MRSA (methicillin-resistant Staphylococcus aureus), drug-resistant tuberculosis (MDR-TB and XDR-TB), and CRE bacteria, resistant to last-resort carbapenem antibiotics. A study published in The Lancet in 2022 estimated that bacterial AMR was directly responsible for about 1.27 million deaths worldwide in 2019 and was associated with nearly 5 million deaths.

What Causes Antibiotic Resistance?

Antibiotic resistance is a natural evolutionary process that is accelerated by human misuse. When bacteria are exposed to an antibiotic, the susceptible ones die, but a few that carry random mutations making them resistant survive and multiply, passing on the resistance. Bacteria can also share resistance genes with each other through small DNA rings called plasmids, even between species. The main causes of the problem are the overuse of antibiotics, such as taking them for viral infections like colds; the misuse, such as not completing the prescribed course or taking the wrong dose; self-medication and buying antibiotics without a prescription, which is common in India; the widespread use of antibiotics in livestock, poultry and aquaculture for growth promotion; poor sanitation and infection control in hospitals; lack of clean water; and few new antibiotics being developed. Resistant bacteria spread across the world through travel and trade. The New Delhi metallo-beta-lactamase gene (NDM-1), identified in 2008 in a patient who had been treated in India, is a famous example of a resistance gene which spread globally.

AMR and India

India is among the largest consumers of antibiotics in the world, and also has a high burden of bacterial infections, which makes AMR a big concern. In 2012 medical societies met in Chennai and issued the Chennai Declaration, a set of recommendations for a national policy on antibiotic use, one of India's first major steps. The government launched the National Action Plan on AMR in 2017, covering awareness, surveillance, and rational antibiotic use, and since March 2014 has placed many antibiotics in Schedule H1, which requires a prescription and record keeping by pharmacists. Kerala started Operation AMRITH to curb the unprescribed sale of antibiotics. In 2019 India banned colistin, a last-resort antibiotic, in animal feed. World AMR Awareness Week is observed from 18 to 24 November.

What You Can Do

Take antibiotics only when prescribed by a qualified doctor, follow the dose and the duration exactly, do not share or save leftover antibiotics, do not demand antibiotics for colds and coughs, wash hands regularly, keep vaccinations up to date, practise good food and water hygiene, and dispose of unused medicines safely.

Challenges and Future

The discovery of new antibiotics has slowed down because it is expensive and profits are low. Promising approaches include new antibiotic classes, bacteriophage therapy, which uses viruses that infect bacteria, rapid diagnostic tests that tell whether an infection is bacterial, antibiotic stewardship programmes in hospitals, vaccines that prevent infections and the use of artificial intelligence to find new drugs. AI discovered a new antibiotic called halicin in 2019.

Quick Facts

- Alexander Fleming discovered penicillin in 1928.
- Antibiotics do not work on viruses because viruses are not cells and lack the targets that antibiotics attack.
- Antibiotic resistance is caused by overuse and misuse of antibiotics, which lets resistant bacteria survive and multiply.
- The Chennai Declaration of 2012 was an early Indian effort to curb antibiotic misuse.
- India placed many antibiotics in Schedule H1 in March 2014 to restrict over-the-counter sales.